

MATILDE MENINI

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PROFILE

MSc student in Statistical Sciences (University of Bologna) with a BSc in Biotechnology and research experience in bioinformatic and statistical analysis. Seeking internships and thesis projects in bioinformatics, deep learning, epidemiology and biomedical data analysis.

EDUCATION

MSc in Statistical Sciences (English-taught programme)

Sep 2025 – present

University of Bologna

- Machine learning, statistical learning, data mining, survival analysis, epidemiology, biodemography, deep learning, data visualization, network analysis, text analysis, LLMs.
- Planned: course in “*Distributed Systems for Data Science*” carried out through *Una Europa exchange programme* at the University of Zurich (UZH) in fall 2027.

BSc in Biotechnology (English-taught programme)

Sep 2022 – Sep 2025

University of Bologna

- Statistical modelling, bioinformatics, genomics, epidemiology, experimental design and data management.
- Hands-on wet lab coursework, including design and execution of laboratory assays.

RESEARCH & INTERNSHIPS

Intern, Infectious Disease Surveillance – Istituto Superiore di Sanità (Rome)

February 2027 – June 2027 (planned)

Planned activities:

- Statistical analyses for epidemiological studies, with a focus on respiratory disease surveillance.
- Application of regression models, supervised statistical learning and survival analysis to assess health outcomes and forecast epidemiological trends.
- Collaboration with the research team to interpret results in support of public health decision-making.

Intern, Bioinformatics and Metagenomics – University of Bologna

Feb 2024 – May 2024

- Designed and implemented a parallelized statistical pipeline in R to analyse a dataset of MAGs.
- Applied multivariate methods (PCoA, PERMANOVA) and non-parametric tests to assess significant genomic variation across ecological niches and bacterial taxonomic levels.
- Applied multiple testing correction (FDR) and modelled gene presence/absence in relation to genome completeness.
- Contributed to a large scale phylogenomic study using bioinformatics tools (VirSorter2, DRAM-v, IQ-TREE, happi, PPanGGOLiN, DefenseFinder).

SKILLS

Programming: Python, R, SQL, Bash, C++

Data analysis: statistical modelling, machine learning, data mining, visualization, experimental design

Certifications: Data Visualization and Data Manipulation in SAS

Languages: Italian (native); English (C1); French (DELF B2)

Soft skills: teamwork, adaptability to different academic and work environments

RESEARCH

Currently writing a research project on diversity of microbial communities (University of Bologna) “Statistical analysis of genomic and functional diversity in Terriglobia across environments”.

OTHER WORK EXPERIENCE

Waitress / Beach Attendant – Beach club “La Playita”, Lavagna (GE)

Jun 2026 – Sep 2026

Customer service, teamwork and working under pressure.

Receptionist / Guest Assistant – Hotel “Celeste”, Sestri Levante (GE)

Jun 2025 – Sep 2025

Managed check-in, bookings and guest requests, communicating with international guests.

Waitress / Beach Attendant – Seaside resort “Bagni Aldebaran”, Lavagna (GE)

Jun 2024 – Sep 2024

Table service and customer assistance in a fast-paced seasonal environment.